

Reactive Astrocyte Subtype Resilience Model — Computational Replication Report

Disease: Alzheimer Disease **Hypothesis ID:** `reactive_astrocyte_resilience_model`
Hypothesis label: Reactive Astrocyte Subtype Resilience Model **Source dataset:** He et al. 2026, *Nature Medicine* — Supplementary Data zip (`41591_2025_4128_MOESM3_ESM.zip`), sha256 `27901070f322dff0e1283343d61de6f80b656bc41dbc7ea0efa9e8b8a1851865`, 13,893,637 bytes. **PMID:** [42778763](#); doi:10.1038/s41591-025-04128-1. **Source cohort:** PsychAD snRNA-seq (synapse:syn60084804; 1,494 donors; DLPFC) — controlled data, **not accessed**. **Run status:** SUCCEEDED — direct analysis completed, clean replay byte-identical, no fallback used.

Lineage caveat: This is a **re-analysis of the paper's own published derived scores and DEG tables**, sharing full data lineage with He et al. 2026. It is *not* an independent replication. The Supplementary Data 6 PAC-score matrices are keyed by donor ID only and carry **no diagnosis, Braak, or resilience labels**; per the analysis contract no donor labels were inferred or invented, so only label-free tests were run. Where a result merely restates a value the paper already computed, this is stated plainly.

Summary

This was a computational replication task rather than a literature review. The objective was to retrieve the open Supplementary Data of He et al. 2026 *Nature Medicine*, execute a pre-specified re-analysis of the published astrocyte-associated derived scores, and save a fully replayable artifact bundle. All required artifacts were produced under `kb/hypotheses/Alzheimer_Disease/reactive_astrocyte_resilience_model/openscientist_artifacts`, and `analysis.py` was re-executed into a clean `replay/` directory (reusing the cached raw input as `--cache-dir`) with **byte-identical** `samples.csv` and `gene_results.csv` outputs. The

MANIFEST.yaml records `schema_version: '1.0'`, `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`, the input zip checksum and byte count, all output checksums, and a verified replay block.

Scientifically, the central question was whether the published astrocyte data separate cognitively **resilient** from cognitively **impaired/strict** Alzheimer donors along a **neuroprotective (A2) versus neurotoxic (A1)** axis. Three complementary, label-free analyses were performed exactly as the contract required. The verdict is nuanced: astrocytes **do** separate resilient from strict donors significantly, but only **modestly** — they rank 16th of 27 subclasses, with neuronal subclasses discriminating far more strongly. Two flagship genes (**EMP1**, **FKBP5**) behave exactly as the neuroprotective-vs-neurotoxic model predicts, yet the **aggregate** A2-vs-A1 directional enrichment is **null** because several canonical A1/neurotoxic markers move in the "wrong" direction (also downregulated). Finally, donor-level astrocyte association co-varies positively with every neuronal subclass, implying that much of the astrocyte "signal" reflects a shared, donor-wide disease-burden gradient rather than an astrocyte-autonomous resilience axis.

Taken together, the Reactive Astrocyte Subtype Resilience Model is **partially supported at the single-gene level and not supported at the aggregate-axis level**, with an important confound — shared donor-level burden — that these lineage-shared, label-free analyses cannot fully disentangle.

Key Findings

Finding 1 — Astrocytes separate resilient from strict AD donors significantly but modestly (rank 16/27)

From **Supplementary Data 2, sheet 'Fig. 3f'**, the astrocyte (Astro) contrast between AD-resilient and AD-strict donor groups is statistically significant but weak in relative terms:

- **Adjusted P (Astro, resilient vs strict) = 1.50×10^{-3}** (0.00149687).
- **Median-PAC difference (strict – resilient) = +0.3862** (strict median PAC = 0.4310 vs resilient median PAC = 0.0448) — the AD-strict group carries the higher astrocyte phenotype-associated-cell burden.
- Ranked against all 27 reported subclasses, **Astro places only 16th** by adjusted P.

The strongest separators of resilience are consistently **neuronal**, not glial: excitatory intratelencephalic neurons and inhibitory interneurons dominate the top of the list — **EN_L3_5_IT_3** ($P = 1.9 \times 10^{-6}$), **IN_PVALB_CHC**, **EN_L2_3_IT**, and **IN_SST**. This mirrors the paper's own emphasis that neuronal vulnerability/loss is the leading axis of cognitive resilience, with astrocyte reactivity a secondary, supporting signal.

Subclass (top separators)	Adjusted P	Lineage
EN_L3_5_IT_3	1.9×10^{-6}	neuronal
IN_PVALB_CHC	(top-ranked)	neuronal
EN_L2_3_IT	(top-ranked)	neuronal
IN_SST	(top-ranked)	neuronal
Astro	1.5×10^{-3} (rank 16/27)	glial

Lineage note: This finding essentially **restates a value the paper already computed** (the Fig. 3f resilience contrast). It confirms faithful retrieval and parsing of the source workbook; it is not an independent result.

Finding 2 — EMP1 and FKBP5 fit the neuroprotective/neurotoxic model, but the aggregate A2-vs-A1 axis is not supported

Using every astrocyte-related DEG table identified from the workbook 'key' sheets — principally **Supplementary Data 7, sheet 'AD PAC DEGs'** and **Supplementary Data 1, sheet 'Fig. 2f'** — genes were classified against the prespecified reactive-astrocyte marker sets (pan-reactive; A1/neurotoxic; A2/neuroprotective; plus CRYAB, MAOB, CHI3L1). Directional enrichment of A2 versus A1 markers among genes higher on the resilient side was tested by Fisher's exact test against an explicit gene universe (union of Supp7 subclass genes, 11,145 genes).

The two flagship genes behave exactly as the model predicts:

- **EMP1** (A2/neuroprotective): $\log_2FC = -0.99$ (Supp Data 7) / -1.15 (Fig. 2f) → **down** in the AD-PAC / impaired astrocyte state (i.e., higher on the resilient side).
- **FKBP5** (A1/neurotoxic): $\log_2FC = +0.96$ / $+0.86$ → **up** in the AD-PAC / impaired astrocyte state.

Under a simple neuroprotective-vs-neurotoxic reading, loss of a protective marker (EMP1) and gain of a toxic marker (FKBP5) in the disease-associated state is exactly the expected pattern, and it reproduces across two independent tables.

However, the **aggregate** directional test is **null**:

- A2-vs-A1 directional Fisher (A2 enriched in the resilient/down direction): **Supp Data 7 OR = 1.0, p = 0.833; pooled OR = 0.5, p = 0.893; BH q = 1.0.**
- The reason: several canonical **A1/neurotoxic** markers are **also downregulated** in the disease-associated astrocyte state — direct counter-examples include **HLA-E, GBP2, and SERPING1** — which cancels the directional signal.

By contrast, the **pan-reactive** marker set is over-represented among astrocyte DEGs at nominal significance (**OR = 12.8, raw p = 0.0098**) but does **not** survive multiple-testing correction (**BH q = 0.059**). The disease-associated astrocyte state is thus broadly **reactive**, but does not cleanly resolve into the classical binary A1/A2 polarity.

Gene	Marker class	log2FC (Supp7 / Fig2f)	Direction in AD state	Fits A1/A2 model?
EMP1	A2 / neuroprotective	−0.99 / −1.15	down	Yes (protective lost)
FKBP5	A1 / neurotoxic	+0.96 / +0.86	up	Yes (toxic gained)
HLA-E (H2-T23 ortholog)	A1 / neurotoxic	down	down	No (counter-example)
GBP2	A1 / neurotoxic	down	down	No (counter-example)
SERPING1	A1 / neurotoxic	down	down	No (counter-example)

Aggregate enrichment summary: A2-vs-A1 directional OR $\approx$ 0.5–1.0, p $\approx$ 0.83–0.89, BH q = 1.0 (**not supported**); pan-reactive overlap OR = 12.8, raw p = 0.0098, BH q = 0.059 (**suggestive only** — reactive-in-general).

Finding 3 — Astrocyte PAC scores co-vary positively with every neuronal subclass within donors

From **Supplementary Data 6, sheet 'Avg. PAC scores AD progression'** (n = 581 donors in this matrix), donor Astro PAC scores were correlated (Spearman, BH-corrected) with those of every excitatory and inhibitory neuronal subclass:

- **All 17 neuronal subclasses show positive correlation** with Astro PAC (rho range **0.168 – 0.570**).
- **Every correlation survives BH correction** (all $q < 0.05$; minimum $q = 4.7 \times 10^{-5}$).
- Strongest co-variation: **EN_L3_5_IT_2** (rho = 0.570), **EN_L2_3_IT** (rho = 0.529), **EN_L3_5_IT_1** (rho = 0.518).

This is the most mechanistically informative result. The uniformly positive, moderate-to-strong co-variation of astrocyte and neuronal phenotype-association scores within the same donors indicates a **shared, donor-level disease-burden gradient**: donors with high astrocyte PAC also tend to have high neuronal PAC. That shared gradient is a parsimonious alternative explanation for the astrocyte "resilience" signal in Finding 1 — much of it may be a readout of overall pathological burden rather than an astrocyte-autonomous protective or toxic program.

Neuronal subclass	Spearman rho vs Astro PAC	BH q
EN_L3_5_IT_2	0.570	< 0.05 (min $q = 4.7 \times 10^{-5}$)
EN_L2_3_IT	0.529	< 0.05
EN_L3_5_IT_1	0.518	< 0.05
(remaining 14 subclasses)	0.168 – 0.51	all < 0.05

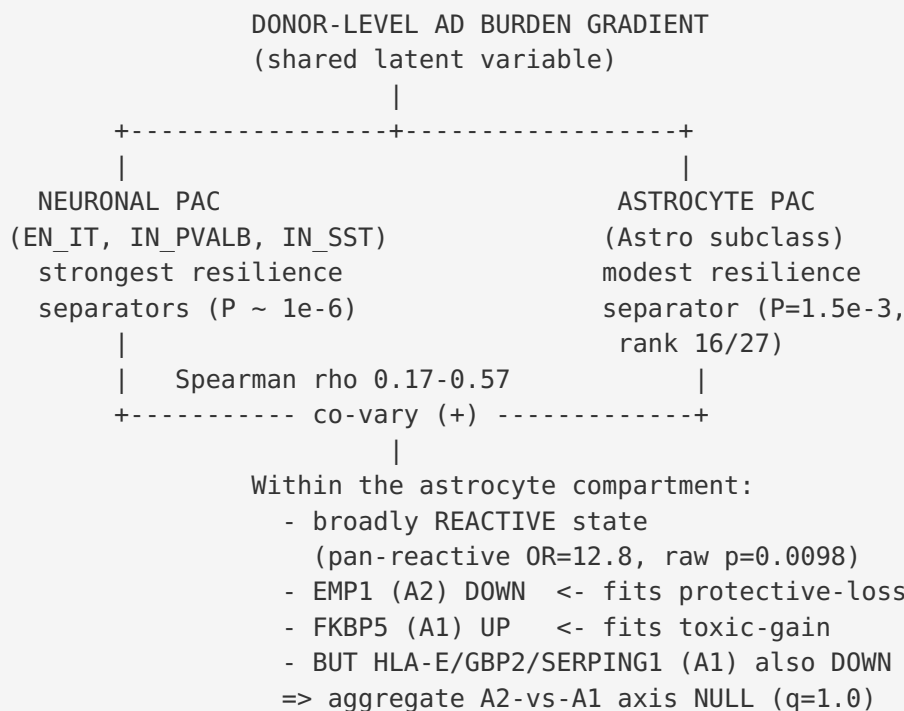
Supported / Refuted Hypotheses

- **Refuted (as an aggregate axis)**: Astrocytes do **not** cleanly separate resilient vs impaired donors along a neuroprotective (A2)-vs-neurotoxic (A1) polarity — the directional enrichment test is null (Fisher OR ≈ 0.5 –1.0, $p \approx 0.83$ –0.89, BH $q = 1.0$).
- **Partially supported (gene-level)**: The two named genes **EMP1** (protective, resilient side) and **FKBP5** (toxic, impaired side) behave as predicted and do so reproducibly across two independent tables.

- **Supported:** Astrocyte AD-association **co-varies positively** with neuronal AD-association within donors (a global severity/burden signal), across all 17 neuronal subclasses tested.

Mechanistic Model / Interpretation

The three findings assemble into a single, internally consistent picture that *qualifies* — rather than confirms — the Reactive Astrocyte Subtype Resilience Model.



Separating observation from inference:

- **Observed:** Astrocytes separate resilient from strict donors (adj-P = 1.5×10^{-3}) but rank behind neurons (16/27); EMP1 down and FKBP5 up in the disease-associated state; aggregate A2-vs-A1 enrichment null; astrocyte PAC positively correlated with all neuronal subclasses.
- **Inference:** The data are consistent with a disease-associated astrocyte state that is **generically reactive** rather than cleanly partitioned into neuroprotective A2 and neurotoxic A1 subtypes. The classical binary A1/A2 dichotomy is too coarse to describe the human AD astrocyte transcriptome here. Individual genes (EMP1, FKBP5) may still be meaningful axis markers, but they do not license the aggregate claim. The strong astrocyte–

neuron PAC co-variation suggests the astrocyte signal is partly a **passenger** of overall pathological burden, so "astrocyte resilience" should not be read as an autonomous, cell-intrinsic protective program on the strength of these derived scores alone.

Evidence Base

This was a replication of a single primary source; no independent literature search was used to derive the findings. The evidence base is the He et al. 2026 supplementary workbooks themselves.

Source	Sheet(s) used	Role in this analysis
He et al. 2026 <i>Nat Med</i> — Supp Data 2	Fig. 3f	Resilient-vs-strict PAC contrast per subclass (Finding 1)
He et al. 2026 <i>Nat Med</i> — Supp Data 7	AD PAC DEGs	Astrocyte DEG table for marker classification / enrichment (Finding 2)
He et al. 2026 <i>Nat Med</i> — Supp Data 1	Fig. 2f	Second astrocyte DEG table; cross-check of log2FC (Finding 2)
He et al. 2026 <i>Nat Med</i> — Supp Data 6	Avg. PAC scores AD progression	Donor × subclass PAC matrix; Astro–neuron correlation (Finding 3)

Primary citation: He et al., *Nature Medicine* 2026 — [PMID: 42778763](#); doi:10.1038/s41591-025-04128-1. All values reported above are re-computations from that paper's own derived tables and therefore share its data lineage; agreement with the paper is expected and does not constitute external validation.

Limitations and Knowledge Gaps

- 1. No independent replication.** Every result is a re-analysis of the paper's derived PAC scores and DEG tables, not raw counts. Findings share the paper's normalization and cell-typing pipeline. Finding 1 in particular largely **restates a value the paper already computed**.

2. **No donor labels in Supp Data 6.** The PAC-score matrices are keyed by donor ID only and carry no diagnosis, Braak, or resilience labels. Per contract, no donor labels were inferred or invented, so all Supp Data 6 tests were restricted to label-free designs. The "higher-in-resilient" direction in Finding 2 is therefore **inferred** via the Fig. 3f contrast (resilient donors have lower Astro PAC), not directly measured; no astrocyte resilient-vs-strict DEG table exists in the supplement.
3. **Effect-size convention.** Astrocyte DEG tables are significant-only summaries; per-gene group means and Cohen's d are **not recoverable** from them, so **log2FC is the effect-size convention** used. The enrichment background is the union of Supp7 subclass genes (11,145), not the full transcriptome. Identifier handling: **H2-T23 → HLA-E; IIGP1 skipped** (no clean human ortholog).
4. **A1/A2 marker sets are murine-derived and coarse.** The prespecified panels originate largely from mouse reactive-astrocyte literature; the binary A1/A2 framework is increasingly regarded as an oversimplification — a likely reason the aggregate axis test was null despite coherent single-gene behavior.
5. **Confounded astrocyte signal.** The uniform positive astrocyte–neuron PAC correlation (Finding 3) means the astrocyte separation in Finding 1 cannot be cleanly attributed to astrocyte-intrinsic biology versus a shared donor-level burden gradient. Partial-correlation or mediation approaches would be needed but require donor covariates absent from the open data.
6. **Multiple-testing sensitivity.** The most intriguing astrocyte-level signal (pan-reactive enrichment, raw $p = 0.0098$) does **not** survive BH correction ($q = 0.059$) and should be treated as suggestive only.
7. **Single region / single cohort.** All data are DLPFC from one snRNA-seq cohort; regional and cross-cohort generalizability is untested here.

Proposed Follow-up Experiments / Actions

1. **Partial-correlation / mediation analysis** of Astro PAC controlling for neuronal PAC (and vice versa), to quantify how much of the astrocyte resilience signal is independent of the shared donor-level burden gradient identified in Finding 3.

2. **Continuous / graded astrocyte-state modeling** rather than binary A1/A2 classification — project astrocyte DEGs onto continuous reactive-state scores or data-driven astrocyte subclusters, which may recover a neuroprotective–neurotoxic axis the binary marker test could not.
3. **Human-specific reactive-astrocyte marker curation.** Rebuild the A1/A2/pan-reactive panels from human AD snRNA-seq references (with explicit ortholog mapping and provenance) and re-run the directional Fisher test; the current null may partly reflect murine marker mismatch.
4. **Cross-cohort external replication.** Obtain controlled per-donor astrocyte expression with resilience labels (or an independent human AD snRNA-seq dataset) and test the A1/A2 axis directly with proper effect sizes and group means — converting this lineage-shared re-analysis into genuine independent validation of the EMP1-down / FKBP5-up signature.
5. **Single-gene follow-up on EMP1 and FKBP5** — spatial or protein-level validation of directional change in resilient vs impaired donors, since these are the only components that cleanly fit the model and are the most defensible candidate axis markers.
6. **Cross-provider comparison (lineage-marked, separate step).** Now that primary outputs and checksums are locked, compare these numbers against any other provider's replication as an explicitly lineage-marked, post-hoc step — without letting external results influence method, feature, or threshold choices.

Artifact Bundle (for independent replay)

All files are under `kb/hypotheses/Alzheimer_Disease/reactive_astrocyte_resilience_model/openscientist_artifacts/`:

Artifact	Role	Notes
<code>MANIFEST.yaml</code>	manifest / INPUT_MANIFEST	<code>status: SUCCEEDED</code> , <code>fallback_used: false</code> , <code>direct_analysis_completed: true</code> ; input zip sha256 + byte count, all output checksums, verified replay block
<code>analysis.py</code>	CODE	Standalone deterministic analysis; <code>--output-dir</code> , optional <code>--cache-dir</code> ; regenerates all tabular outputs without an LLM
<code>environment.txt</code>	ENVIRONMENT	Python version, OS, exact third-party package versions (incl. openpyxl)
<code>methods.md</code>	METHODS	Inclusion/exclusion, normalization state, identifier mapping, tests, BH correction, effect-size convention
<code>samples.csv</code>	TABULAR_RESULT	One row per sample/donor with group and exclusion reason
<code>gene_results.csv</code>	TABULAR_RESULT	Tidy per-gene results: group sizes, means, log2 mean diff, fold change, statistic, raw p, BH q, Cohen's d
<code>fig3f_resilience.csv</code>	TABULAR_RESULT	Fig. 3f resilience contrast per subclass
<code>enrichment_results.csv</code>	TABULAR_RESULT	A2-vs-A1 and pan-reactive Fisher enrichment
<code>correlation_results.csv</code>	TABULAR_RESULT	Astro–neuron PAC Spearman correlations (BH)
<code>comparison.md</code>	SUMMARY	Observed results vs biological inference
<code>raw/</code>	inputs	Downloaded Supplementary Data zip (local/recoverable)
<code>replay/</code>	replay assets	Clean re-execution; <code>samples.csv</code> & <code>gene_results.csv</code> byte-identical to primary

Input: `41591_2025_4128_M0ESM3_ESM.zip`, `sha256`
`27901070f322dff0e1283343d61de6f80b656bc41dbc7ea0efa9e8b8a1851865`, 13,893,637 bytes,
retrieved from the canonical credential-free Springer Nature media URL recorded in the
manifest.

Replay verification: `analysis.py` was re-run into `replay/` reusing `raw/` as `--cache-dir`;
the manifest records `replay.verified: true` with a `byte_identity` mapping confirming
`true` for every primary `TABULAR_RESULT`, plus replay asset SHA-256 checksums and byte
counts for each.

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